

Weather Data ETL Pipeline

Project Overview

1. Data Source

The data for this project was sourced from the **Open Weather API** (openweathermap.org/api), a free, publicly available weather data provider. Using a personal API key generated through Open Weather, live weather data was retrieved for six selected cities in Nigeria: Abuja, Lagos, Kano, Lokoja, Benue, and Oyo.

2. ETL Process

This project followed the standard ETL (Extract, Transform, Load) framework used in data analytics workflows:

Extract — raw weather data was pulled directly from the Open Weather API using Python's requests library. A request was sent for each of the six cities, and the API returned real-time weather details in JSON format.

Transform — the raw JSON data was cleaned and restructured using Pandas. Relevant fields (city name, temperature, humidity, weather condition, wind speed, and date/time) were extracted from the messy JSON response and organized into a structured table (Data Frame). Column names were renamed for clarity, and data types were checked to ensure consistency.

Load — The final cleaned dataset was saved as a CSV file, making it ready for further analysis, sharing, or reuse without needing to re-call the API.

3. Tools Used

- Python — core programming language
- Jupyter Notebook (via Anaconda) — development environment
- Requests library — for connecting to and pulling data from the API
- Pandas — for data cleaning, transformation, and structuring
- Open Weather API — data source
- CSV format — for storing the final dataset

4. Steps Taken

1. Created a free Open Weather account and generated a personal API key
2. Wrote Python code to extract live weather data for six Nigerian cities
3. Converted the API's raw JSON response into a clean, structured format using Pandas
4. Renamed columns and verified data types for consistency and readability
5. Saved the final cleaned dataset as a CSV file for analysis
6. Performed basic analysis on the dataset — comparing temperatures, identifying humidity extremes, and reviewing weather conditions across cities

5. Key Findings

- Kano recorded the highest temperature among the six cities (35.44°C)
- Oyo recorded the lowest temperature (24.56°C)
- Despite having the lowest temperature, Oyo recorded the highest humidity, while Kano — the hottest city — recorded the lowest humidity, suggesting an inverse relationship between temperature and humidity in this dataset
- Weather conditions varied across the six cities, reflecting Nigeria's diverse regional climate patterns

Prepared by Gabriel O. Samuel — Analyst Lab Africa Data Analyst Intern Program